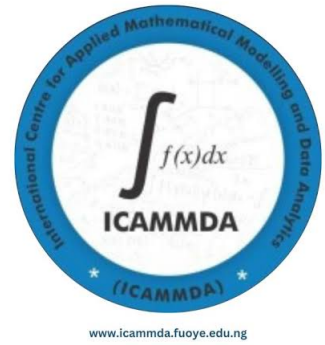




Federal University Oye Ekiti (FUOYE)



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PREDICTING CASSAVA MOSAIC OUTBREAK IN IBADAN USING ENVIRONMENTAL AND CLIMATIC FACTORS DATA: MACHINE LEARNING ALGORITHMS APPROACH

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Abstract

Cassava mosaic disease (CMD), caused by cassava mosaic begomoviruses (CMBs), remains a significant concern for stakeholders, particularly small-scale farmers and agribusinesses in Nigeria. This study aimed to investigate the trends of CMD outbreaks and assess the influence of environmental factors (temperature, humidity, and rainfall) on disease incidence. The research employed Random Forest modeling with 500 trees to analyze the relationships between environmental variables and disease occurrence. Environmental data and disease incidence were collected and analyzed using both correlation analysis and machine learning techniques. The investigation revealed complex relationships between environmental factors and CMD incidence, with the Random Forest model explaining 5.22% of the variance in disease incidence. Correlation analysis showed weak positive associations between temperature and disease incidence ($r = 0.14, 0.13$), while humidity and rainfall exhibited negligible negative correlations. Notably, a strong temporal correlation ($r = 0.82$) between one-month and three-month disease incidence indicated high stability in disease patterns once established. The findings suggest that environmental variables alone are not primary drivers of CMD development, challenging conventional assumptions about weather-disease relationships. This research contributes to understanding CMD epidemiology and suggests that effective disease management strategies should extend beyond weather-based forecasting to include comprehensive approaches incorporating vector monitoring and resistant variety deployment. Keywords: Cassava mosaic disease, Random Forest modeling, environmental factors, disease prediction, begomoviruses.

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INTRODUCTION

Cassava, *Manihot esculenta* Crantz, is an important staple food that serves as an affordable source of carbohydrates for over 800 million people across Africa (FAO, 2013). It is globally ranked the fourth most important staple food crop after rice, wheat, and maize (Rey and Vanderschuren, 2017). It grows well in dry regions without the need for fertilizer and is estimated to feed nearly one billion people worldwide. It is cultivated by local small-scale farmers and international agricultural conglomerates employing vertically integrated production models (Chaiyana et al., 2024). While playing a pivotal role in the global food chain, the sustained production of cassava faces multifaceted challenges, including climate change, pest infestation, soil erosion, fecundity issues, yield reduction, labor concerns, and inequitable cost/sell ratios (Ukoha, 2020). In Nigeria, cassava provides more than half of the daily calorie requirement for people across various ethnic groups (Eni et al., 2020). It is considered an important food security crop, mainly because of the ease of production and processing, as well as its ready to eat product referred to as garri which can be stored for up to 12 months at room temperature (Adu, 2020). Besides its importance for food, cassava is also an industrial raw material and thus a potential source of income to farmers and the country at large. Although Nigeria is the largest producer of cassava in Africa, with about 60 million tonnes produced annually (FAO, 2017), the five to 10 tonnes per hectare tuber yield common in Nigeria is much lower than the average tuber yield of 25 tonnes per hectare obtained in other cassava growing regions around the world (FAO, 2017). One major biotic constraint to cassava production is its susceptibility to cassava mosaic disease (CMD), a viral disease which causes annual tuber yield losses estimated at USD 1.9 to 2.7 billion (Sastry and Sastry, 2013).

CMD is caused by a group of viruses commonly referred to as cassava mosaic begomoviruses (CMBs), belonging to the genus *Begomovirus* in the family *Geminiviridae* (Alabi et al., 2015). One of the primary concerns among stakeholders, encompassing small-scale farmers and agribusinesses, revolves around the prevalence of plant pathogens, particularly the Cassava Mosaic Disease (CMD) (Minato et al., 2019). CMD stands out as the most widespread crop disease affecting cassava globally (Herrera Campo et al., 2011), causing substantial crop damage and exponential yield reductions over time (Kalyebi et al., 2021). The CMBs are transmitted through use of infected cassava cuttings as planting materials

and by whitefly vectors belonging to the *Bemisia tabaci* complex (Elfekih et al., 2021). The whitefly easily transmits the virus over large areas as it feeds on the phloem of plants and transmits the virus through its infected saliva (Aimone et al., 2021). Characteristic symptoms of CMD vary from mosaic patterns on cassava leaves to leaf distortion, vein clearing and stunted growth (Koima, 2018). Several strains of CMBs have been identified to cause CMD in Africa namely African cassava mosaic virus (ACMV) South African cassava mosaic virus (SACMV), East African cassava mosaic virus (EACMV), East African cassava mosaic Cameroon virus (EACMCV), East African cassava mosaic Zanzibar virus (EACMZV), East African cassava mosaic Malawi virus (EACMMV), and East African cassava mosaic Kenya virus (EACMKV). Recently, two species have also been described: African cassava mosaic Burkina Faso virus (ACMBFV) and Cassava mosaic Madagascar virus (Soro et al., 2021). Nationwide surveys conducted in Nigeria for the assessment of CMD and CMB in 2002 and 2003 revealed that ACMV, EACMV and EACMCV are the predominant CMBs responsible for CMD (Alabi et al., 2024). CMBs have also been identified in other plant species besides cassava including *Laportea* (*Fluerya*) *aestuans* (*Urticaceae*), *Senna occidentalis* (L.) Link, *Manihot glaziovii*, *Combretum confertum* (Benth.) M.A. Lawson, *Ricinus communis* L., *Leucana leucocephala* (Lam.) De Witt, *Glycine max* (L.) Merr. (Alabi et al., 2008). These non-cassava CMBs hosts can serve as a reservoir for the spread of CMBs through whitefly transmission. Detecting and assessing the prevalence of CMD in cassava crops is a crucial step for evaluating the disease's extent (Kalischuk et al., 2019). Employing healthy plant materials, controlling pests, and vigilantly monitoring fields for early signs of infection are essential measures in eradicating the virus (Chaiyana et al., 2024).

Machine learning is an application of Artificial intelligence that proves that systems have the ability to learn from previous experience and improve it without being programmed, and it is a technique that explores the analysis and creation of algorithms that can be used to make data predictions (Modu et al., 2017). There are two categories of machine learning algorithms which are supervised and unsupervised learning algorithms: Supervised learning algorithms are a machine learning algorithm that attempts to model relationships and dependencies between the output of the target prediction and the input features such that the output values for new data can be predicted based on those relationships that it has learned from previous data sets, supervised learning uses labelled data sets (Stephen et al., 2021). Unsupervised Learning is a machine learning approach for detecting patterns in datasets with unlabeled or unstructured data points, In this learning approach, an artificial intelligence system gets just the input data and not the associated output data (Naeem et al., 2023). Unsupervised machine learning, unlike supervised learning, does not need the presence of a person to oversee the model (Alloghani et al., 2020).

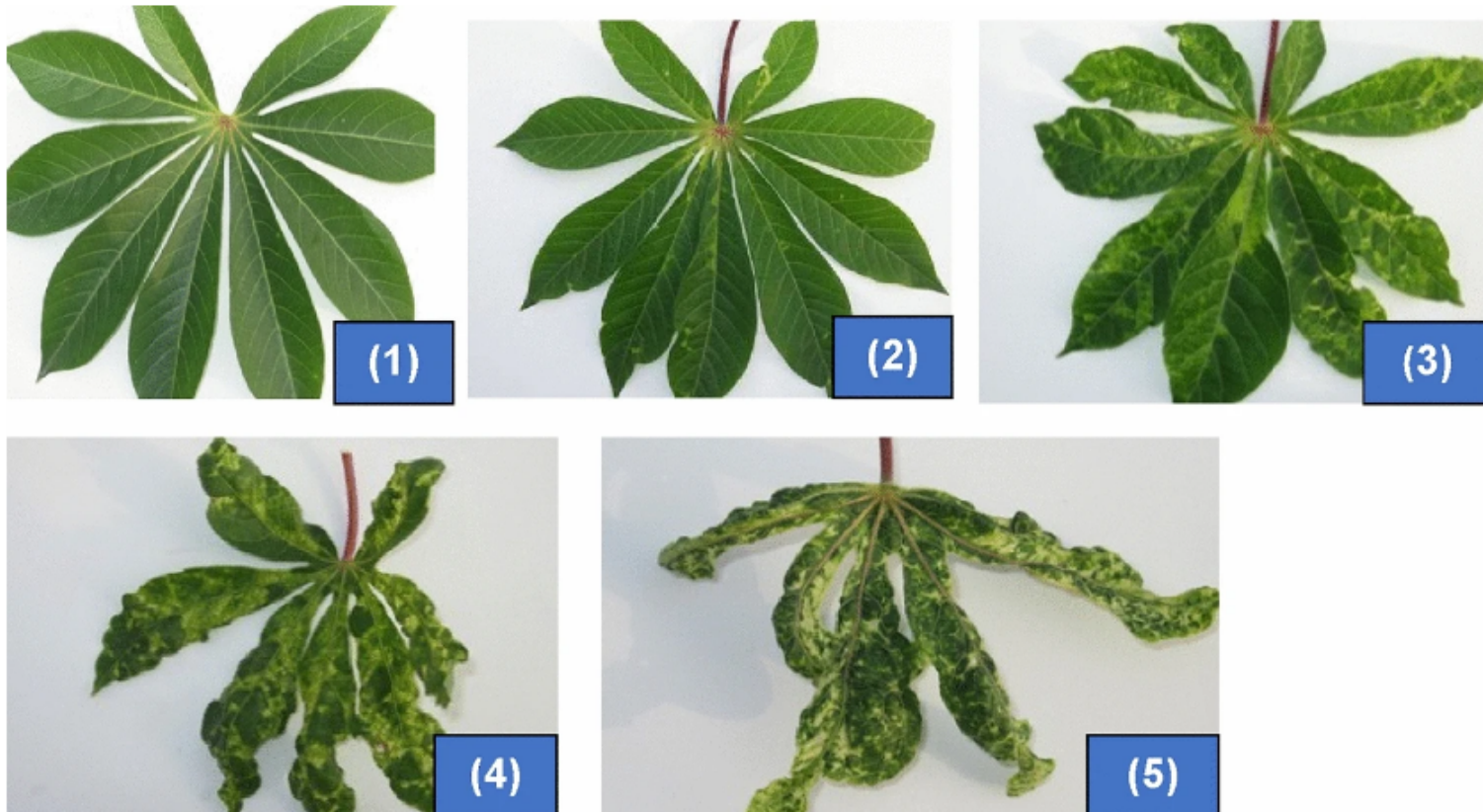


Figure 1.1: Stages of CMD (Olasanmi et al., 2021)

1.1 Aim of the study

The aim of this study is to investigate the trends of cassava mosaic disease outbreaks in Nigeria.

1.2 Research Questions

This study aims to investigate the following research questions:

1. What are the effects of temperature, rainfall and humidity on cassava mosaic disease incidence?
2. What are the correlations among temperature, rainfall, humidity and cassava mosaic disease incidence?

1.3 Objectives of the study

The objectives are to:

1. Access the effect of temperature, humidity and rainfall on cassava mosaic disease incidence.
2. Examine the correlations among temperature, rainfall, humidity and cassava mosaic disease incidence.

1.4 Methodology

- Use Random Forest Regression model to predict the outbreak of cassava mosaic disease.
- Use correlation matrix to investigate the correlation between temperature, rainfall, humidity and cassava mosaic disease incidence.

LITERATURE REVIEW

Research on CMD has advanced significantly, particularly in understanding the disease's biology, epidemiology, and control methods.([Regessa, 2017](#)) The "Disease Control in Cassava Farms: IPM Field Guide for Extension Agents" is a crucial resource that provides practical guidelines for managing CMD through integrated pest management (IPM). It emphasizes the importance of resistant cassava varieties, controlling whitefly populations (which transmit CMBs), and removing infected plants to prevent the disease's spread. These IPM strategies have proven effective across various regions, but they often focus on managing the disease after it has already taken hold, rather than predicting outbreaks in advance.

The study [Eni et al. \(2021\)](#), titled "South West and North Central Nigeria: Assessment of Cassava Mosaic Disease and Field Status of African Cassava Mosaic Virus and East African Cassava Mosaic Virus," offers valuable insights into the prevalence of CMD in Nigeria. Conducted in key cassava-producing areas, the research highlights the widespread presence of both ACMV and EACMV, emphasizing the need for ongoing surveillance and the development of more resilient cassava varieties. This epidemiological data is particularly relevant for informing predictive models aimed at forecasting CMD outbreaks. In recent years, technological advances have enabled new approaches to agricultural disease management.

[Effiong et al. \(2023\)](#) thesis, "Cassava Leaf Disease Classification with Deep Learning," illustrates the application of deep learning to classify cassava leaf diseases, including CMD. By training convolutional neural networks (CNNs) on cassava leaf images, Effiong Blessing Uduakobong achieved high accuracy in disease classification, demonstrating the potential of machine learning in this field. However, transitioning from disease classification to predicting CMD outbreaks represents a significant challenge, requiring the integration of climatic and environmental data into predictive models. Environmental and climatic factors play a pivotal role in the prevalence and severity of CMD outbreaks.

[Eni et al. \(2020\)](#) conducted an extensive assessment in South West and North Central Nigeria, revealing significant regional differences in CMD incidence correlating with varia-

tions in temperature, humidity, and precipitation. Their study underscored that region with climatic conditions favorable for whitefly proliferation experienced higher rates of ACMV and EACMV infections.

Similarly, [Tembo et al. \(2017\)](#) examined CMD incidence and cassava yield performance in Zambia, finding that climatic variables such as increased temperature and rainfall intensity were directly associated with heightened disease prevalence and reduced crop yields. These findings highlight the necessity of incorporating climatic data into CMD management strategies to effectively predict and mitigate outbreaks. ([Farms, 2000](#)) provided a comprehensive guide on disease control in cassava farms, emphasizing the importance of integrated pest management (IPM) approaches that consider environmental factors. Their work suggests that timely monitoring of climatic conditions can facilitate early detection and control of CMD, thereby minimizing economic losses.

The integration of machine learning algorithms presents a promising avenue for enhancing CMD prediction and management. ([Sambasivam and Opiyo, 2021](#)) developed a predictive model utilizing convolutional neural networks (CNNs) to detect and classify cassava diseases, including CMD, from imbalanced datasets. Their model demonstrated high accuracy and efficiency, showcasing the potential of deep learning techniques in processing complex agricultural data for disease prediction. ([Effiong, 2021](#)) further explored the application of deep learning in classifying cassava leaf diseases through image recognition. The study achieved significant success in accurately identifying CMD symptoms, reinforcing the efficacy of machine learning models in facilitating rapid and precise disease diagnosis, which is crucial for timely intervention.

Understanding the relationship between climatic conditions and CMD outbreaks is essential for developing effective predictive models. ([Chikoti et al., 2019](#)) investigated this correlation and found that specific environmental factors, notably elevated temperatures and high humidity levels, significantly contribute to the spread and intensity of CMD. These insights provide a foundational basis for incorporating climatic variables into predictive modeling efforts. ([Legg et al., 2015](#)) expanded on this by examining the biology, epidemiology, and management of cassava virus diseases. Their research highlighted that predictive models integrating environmental and climatic data can substantially improve the accuracy of CMD outbreak forecasts, enabling more proactive and targeted disease management strategies.

Despite the progress in CMD research, several critical gaps remain. While studies like Uduakobong's have shown the effectiveness of machine learning in disease classification, there is a lack of research extending these techniques to predictive modeling for CMD outbreaks. Existing studies often focus on the current distribution and status of CMD, with limited exploration of the relationships between climatic variables and CMD outbreaks.

Moreover, the literature does not adequately quantify the influence of climatic factors such as temperature, humidity, and rainfall on CMD outbreaks. Although some studies acknowledge climate's role in CMD spread, these relationships are not fully explored or utilized in predictive models. Additionally, traditional statistical methods used in existing studies may not capture the complex, non-linear relationships between climatic variables and CMD outbreaks. This highlights the need for more sophisticated machine learning algorithms capable of modeling these intricate interactions to make accurate predictions.

METHODOLOGY

This chapter details the methodological framework adopted for this study to establish a relationship between environmental factors and Cassava Mosaic Disease (CMD) Incidence. The research applied the Random Forest regression modeling machine learning technique that can unpack even complex nonlinear relationships in ecological data. This chapter describes the data collection procedures and methods of data analysis used in addressing the objectives of the research. This research will follow a structured methodology comprising several key steps:

3.1 Study Area

Ibadan (Oyo state, Nigeria) is the largest city in West Africa and the second largest in Africa, with land size covering an area of 240 km². The city is located on geographic grid reference longitude 3°5E, latitude 7°20N ([Filani et al., 1994](#)). The hot season lasts for 2.8 months, from January 14 to April 6, with an average daily high temperature above 33.22°C. The hottest month of the year in Ibadan is March, with an average high of 33.33°C and low of 23.33°C and the cool season lasts for 3.1 months, from June 21 to September 25, with an average daily high temperature below 28.89°C. The coldest month of the year in Ibadan is August, with an average low of 21.67°C and high of 27.78°C ([wea](#)). The rainy period of the year lasts for 9.9 months, from February 2 to November 29, with a sliding 31-day rainfall of at least 12.7 mm. The month with the most rain in Ibadan is September, with an average rainfall of 231.04 mm inches, the rainless period of the year lasts for 2.1 months, from November 29 to February 2. The month with the least rain in Ibadan is January, with an average rainfall of 7.62 mm ([wea](#)).

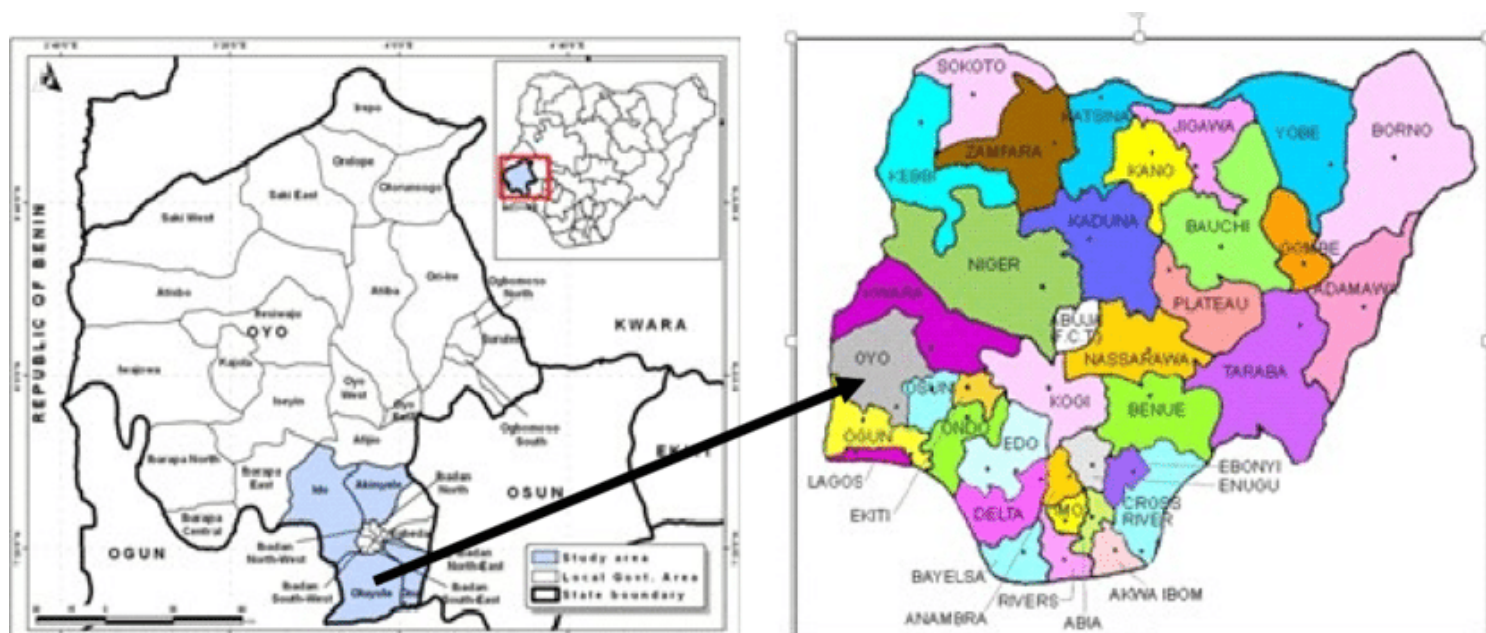


Figure 3.1: Map of Ibadan (Olubanjo, 2016)

3.2 Data

1. Data Sources: The data was obtained from the International Institute for Tropical Agriculture (IITA) and the climatic data was obtained from the Weather and climate site.
2. Data Collection
 - Environmental Variables Data was collected on the following three most relevant environmental variables:
 - Temperature (°C)
 - Relative Humidity (%)
 - Rainfall (mm)
 - Disease Incidence Assessment The incidence of CMD was assessed according to two-time resolutions:
 - One-month incidence
 - Three-month incidence
3. Data Analysis Procedure
 - Data Preprocessing
 - Cleaning: Addressing missing data and outliers.
 - Feature Engineering: Creating additional features that could enhance model performance.
 - Splitting: Dividing the dataset into training and testing sets.
 - Model Training and Validation
 - Training: A model will be developed based on the training dataset.
 - Validation: The performance of the models to be evaluated using cross-validation techniques.
 - Feature importance analysis: It gives the contribution of variables.
4. Model Evaluation:
 - Performance Metrics: Models was evaluated using metrics such as R-squared and mean squared error

3.2.1 Model Development

1. Random Forest Regression: Random Forest is an ensemble learning method that works by constructing a large number of decision trees and then averaging the prediction over

multiple trees. Mathematically, this can be represented as:

$$f(x) = \frac{1}{B} \sum_{b=1}^B f_b(x)$$

Where

$f(x)$ - Random Forest prediction

B -number of trees

$f_b(x)$ - prediction of the tree

x - input variables-temperatures, humidity, rainfall

2. Model Parameters Variable-Random forest parameter No. of trees- 500 Variables per split: 1 Target variable: Disease incidence (month) Predictor variables: Temperature, Humidity, Rainfall

ANALYSIS

This chapter presents the analysis and interpretation of data that were collected on the relationship between environmental factors and the incidence of Cassava Mosaic Disease (CMD). Specifically, the analysis checks on how temperature, rainfall, and humidity influence the development and progression of the disease. The presentation of the findings was done through the use of various statistical analyses such as correlation matrices and Random Forest modeling.

4.1 Analysis of Cassava Mosaic Disease Incidence

4.1.1 Frequency of Incidence After First and Third Month

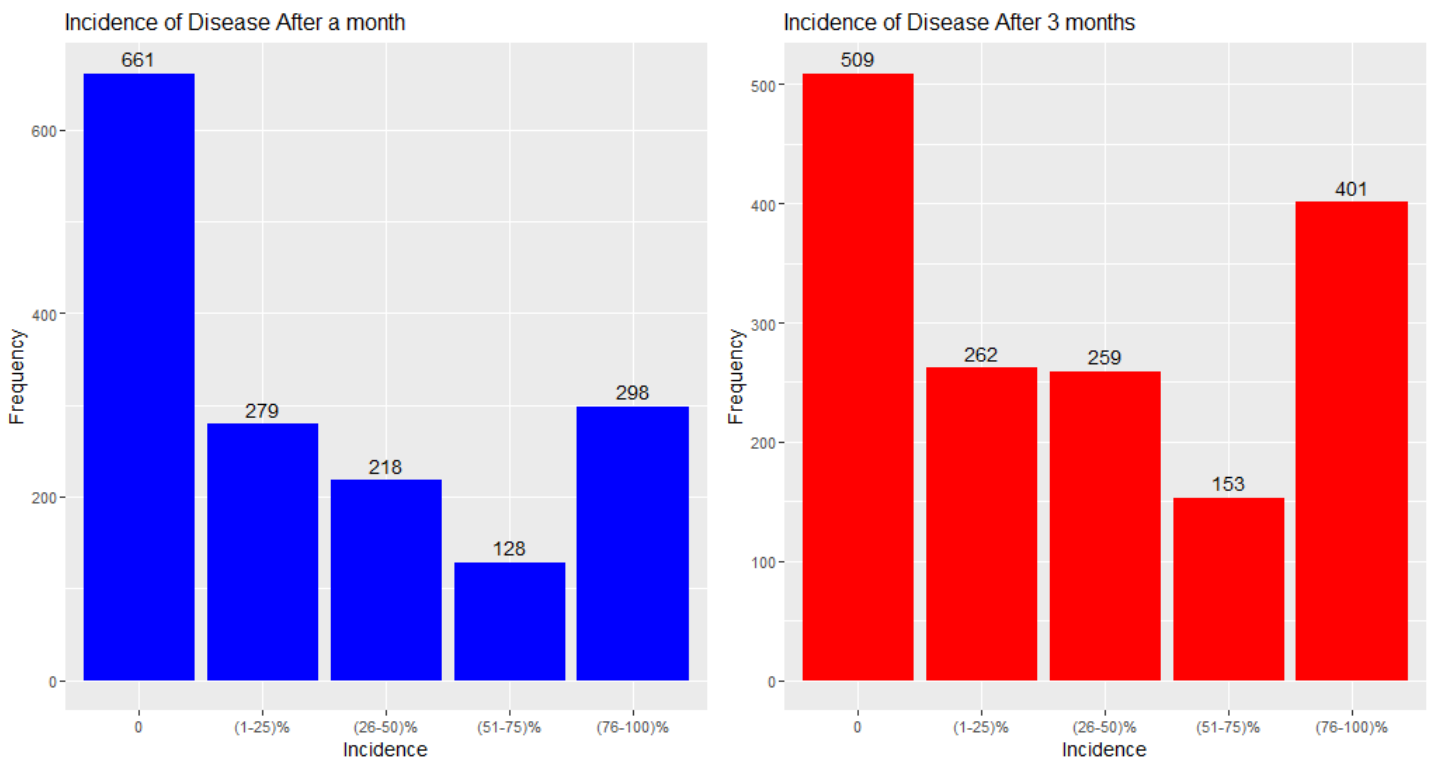


Figure 4.1: Comparizing of Incidence of Disease after a month and 3 months

The analysis of the CMD progression in cassava plants at one and three months, respectively, revealed some interesting patterns. While the plants at one month showed a majority with no incidence of the disease-661 cases were noted, indicating that early manifestation of the disease was minimal. However, in the three-month-old plants, this pattern got reversed, with an overwhelming increase in cases falling into the category of (76-100)% incidence, approximately amounting to 401 cases in number. The data also showed a bimodal distribution, indicating the two distinct responses of plants, one retaining resistance and the other showing a significant development of the disease. It is, therefore, more likely that plant genetics, vector populations, environmental conditions, or even agricultural practices may be considered potential contributors. This three-month period is the most critical stage for the reassessment of the health status of plants and for changing the strategy of management. It therefore requires more intensified monitoring and management of susceptible plants.

4.1.2 Relationship Between Incidence and Climatic Factor

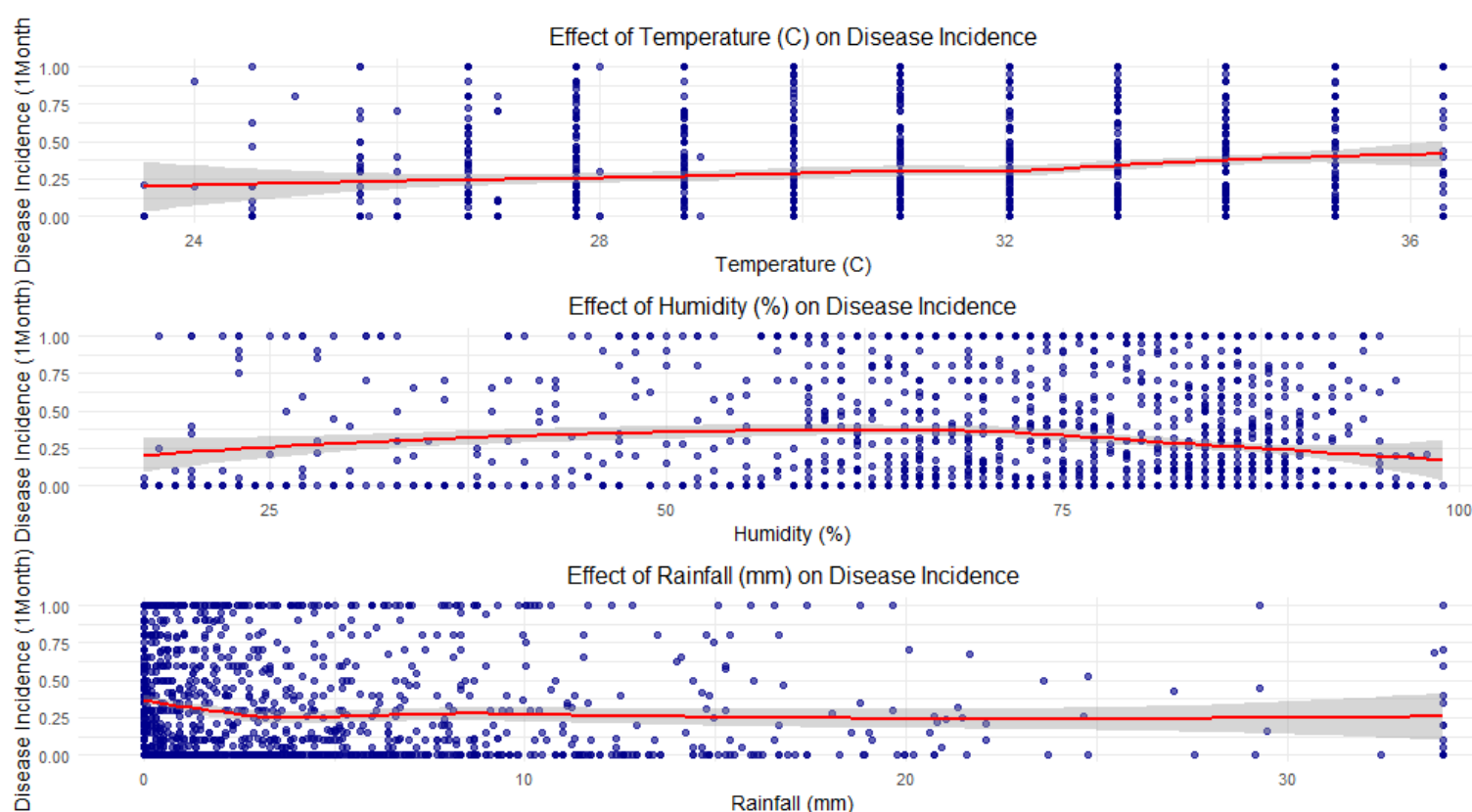


Figure 4.2: Relationship Between Incidence and Climatic Factor

4.2 Random Forest

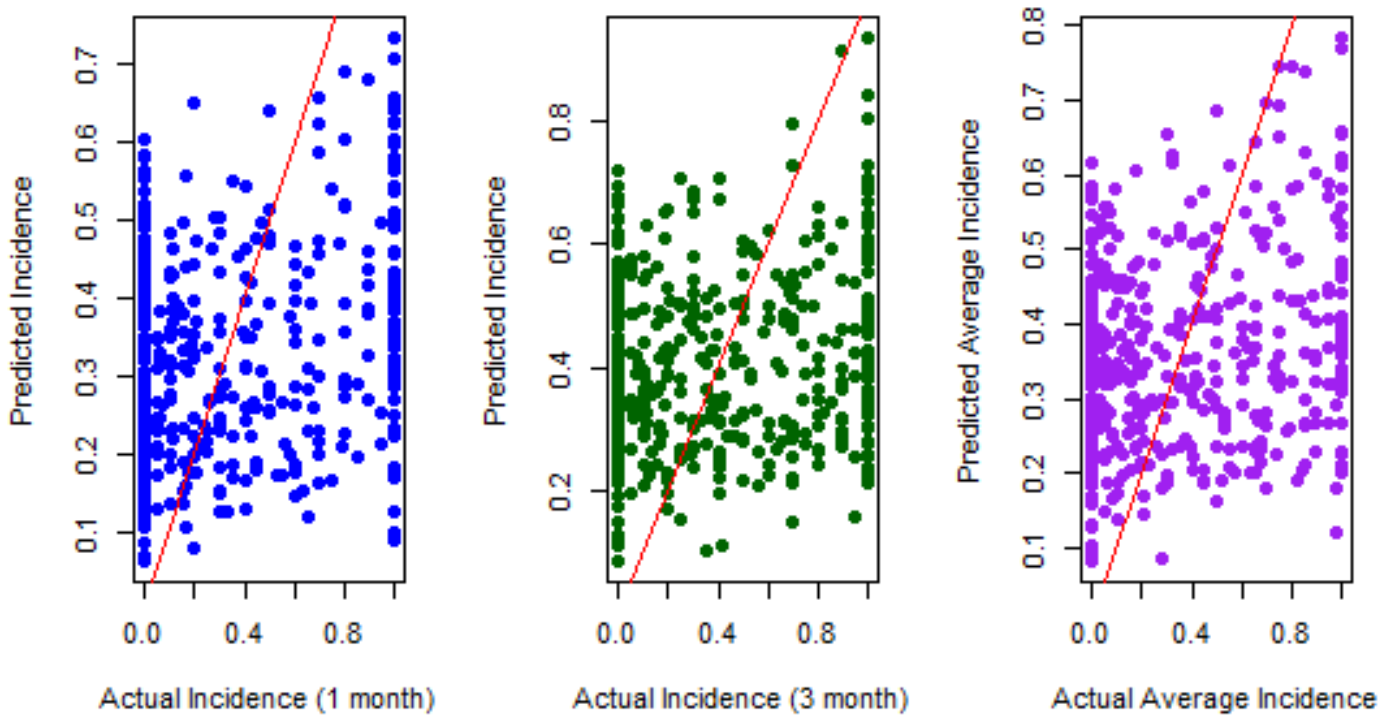


Figure 4.3: Predicted vs Actual

The plots show the relationship between the actual and predicted incidence of cassava mosaic disease for different forecasting periods: 1 month, 3 months, and average incidence.

- **1-Month Incidence Plot (left):** The predicted values are close to the actual values, clustering along the red line (ideal fit). This suggests that the model performs well for short-term predictions despite some deviations.
- **3-Month Incidence Plot (center):** Predictions for the 3-month incidence are more spread out from the red line, indicating that the model's accuracy decreases with the longer forecast period. This may reflect greater uncertainty in capturing disease patterns over a three-month horizon.
- **Average Incidence Plot (right):** The Predicted average incidence shows moderate alignment with actual values, but there is a noticeable spread, indicating variability in the model's performance. The average incidence model appears to capture broader trends rather than exact values.

Overall, the model shows good short-term predictive accuracy, with performance slightly decreasing over longer periods, consistent with typical predictive limitations over time.

4.2.1 Model Evaluation

1. Mean Squared Error (MSE)

The MSE value obtained from our model is 0.128. MSE measures the average squared difference between predicted and actual values, where a lower MSE value indicates a more accurate model. Thus, our MSE result implies that the model has a reasonable error margin but may still have room for improvement in capturing the incidence patterns based on climatic variables.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where:

- y_i - Actual incidence values of cassava mosaic disease.
- $\hat{y}_i$ - Predicted incidence values from the model.
- n - Total number of observations.

2. R-squared (R^2)

The R^2 value obtained is 0.068. This metric explains the proportion of the variance in the cassava mosaic disease incidence explained by the model's climatic predictors (temperature, humidity, and rainfall). An R^2 value closer to 1 signifies a higher predictive power, while values closer to 0 suggest limited predictive relevance. The current R^2 indicates a lower predictive accuracy, suggesting that other variables or model adjustments may be needed to improve the performance.

$$R^2 = 1 - \frac{SSE}{SST}$$

where:

- SSE - Sum of Squared Errors = $\sum_{i=1}^n (y_i - \hat{y}_i)^2$
- SST - Total Sum of Squares = $\sum_{i=1}^n (y_i - \bar{y})^2$, where $\bar{y}$ is the mean of actual values.

4.3 The Correlation Matrix of Climatic Factors and Disease Incidence

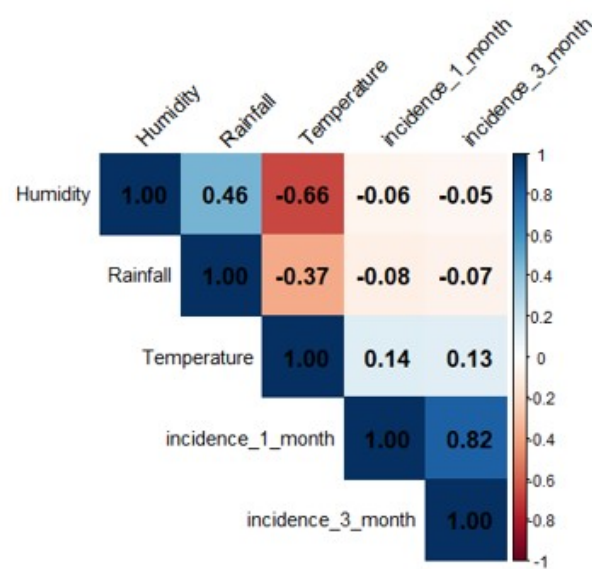


Figure 4.4: The Correlation Matrix of Climatic Factors and Disease Incidence

This plot examined relationships between environmental factors and cassava mosaic disease (CMD) over one-month and three-month periods. Temperature showed weak positive correlations with CMD incidence ($r=0.14, 0.13$), while humidity and rainfall displayed negligible negative correlations. Notably, there was high temporal stability in disease patterns, suggesting that once established, CMD levels remain relatively constant regardless of environmental fluctuations. The study revealed complex interactions among environmental variables, particularly a strong negative correlation between humidity and temperature ($r= -0.66$). These findings indicate that disease management should prioritize prevention over environmental control strategies.

4.3.1 Climatic Factors On the Disease Incidence

The plot below illustrates the importance of different climatic factors (Temperature, Humidity, Rainfall) in predicting cassava mosaic disease incidence using IncMSE and IncNodePurity metrics.

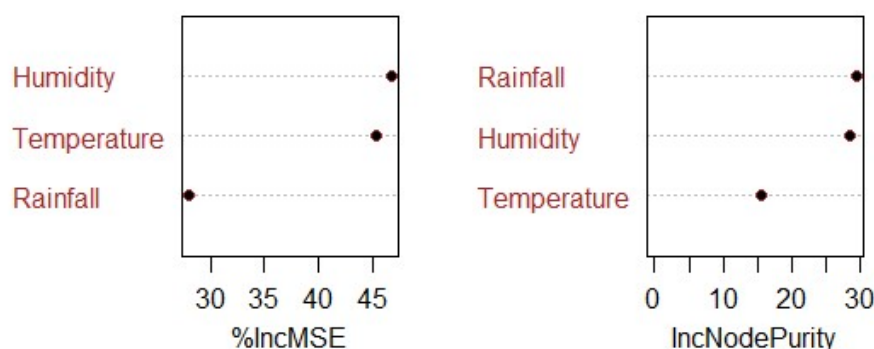


Figure 4.5: Importance of Climatic Factors On the Disease Incidence

1. %IncMSE: This metric reflects how much the mean squared error (MSE) increases when each variable is removed from the model. Higher values indicate a greater impact on prediction accuracy.
 - Temperature has the highest IncMSE, suggesting that it is the most important variable for predicting disease incidence.
 - Humidity shows moderate importance.
 - Rainfall has the lowest IncMSE, indicating it contributes the least to predictive accuracy in this model.
2. %IncNodePurity: This metric indicates each variable's contribution to node purity, with higher values suggesting greater importance in splitting data accurately within the model.
 - Rainfall is the most influential variable by this measure, suggesting it plays a significant role in segmenting the data effectively.
 - Humidity follows, with moderate importance.
 - Temperature has the lowest IncNodePurity score, implying it has less impact on improving model structure through splits.

In summary, while Temperature is key for predictive accuracy (IncMSE), Rainfall is crucial for structuring the model (IncNodePurity), indicating that both factors play important but different roles in predicting disease incidence.

CONCLUSION

The investigation into environmental factors' influence on CMD revealed complex and somewhat unexpected relationships. While temperature showed weak positive correlations with CMD incidence ($r=0.14, 0.13$), both humidity and rainfall demonstrated negligible negative correlations, challenging conventional assumptions about moisture's role in plant disease development. The Random Forest analysis further supported this complexity, explaining only 5.22% of CMD incidence variance, suggesting that environmental variables alone are not primary drivers of disease development.

A notable finding was the high temporal stability in CMD patterns, indicating that established infection levels remain relatively constant despite environmental fluctuations. This, combined with the low explanatory power of weather variables, suggests that CMD epidemiology is governed by more complex factors, potentially including vector populations and host plant resistance. These findings have important implications for disease management, indicating that strategies should extend beyond weather-based forecasting to include comprehensive approaches incorporating vector monitoring and resistant variety deployment.

References

- , . Ibadan Climate, Weather By Month, Average Temperature (Nigeria) - Weather Spark — weatherspark.com. <https://weatherspark.com/y/48855/Average-Weather-in-Ibadan-Nigeria-Year-Round#Figures-Humidity>. [Accessed 25-10-2024].
- Adu, E.O., 2020. Evaluation of yellow flesh cassava (*manihot esculenta crantz*) genotypes for yield, dry matter content, cyanogenic potential total carotenoid content and stability. Ph.D. thesis. University of Cape coast.
- Aimone, C.D., Lavington, E., Hoyer, J.S., Deppong, D.O., Mickelson-Young, L., Jacobson, A., Kennedy, G.G., Carbone, I., Hanley-Bowdoin, L., Duffy, S., 2021. Population diversity of cassava mosaic begomoviruses increases over the course of serial vegetative propagation. *Journal of General Virology* 102, 001622.
- Alabi, O.J., Kumar, P.L., Naidu, R.A., 2024. Cassava and its importance in sub-saharan africa. *Phytopathology News* .
- Alabi, O.J., Mulenga, R.M., Legg, J.P., 2015. Cassava mosaic., in: *Virus diseases of tropical and subtropical crops*. CABI Wallingford UK, pp. 56–72.
- Alabi, O.J., Ogbe, F.O., Bandyopadhyay, R., Lava Kumar, P., Dixon, A.G., d'A. Hughes, J., Naidu, R.A., 2008. Alternate hosts of african cassava mosaic virus and east african cassava mosaic cameroon virus in nigeria. *Archives of Virology* 153, 1743–1747.
- Alloghani, M., Al-Jumeily, D., Mustafina, J., Hussain, A., Aljaaf, A.J., 2020. A systematic review on supervised and unsupervised machine learning algorithms for data science. *Supervised and unsupervised learning for data science* , 3–21.
- Chaiyana, A., Khiripet, N., Ninsawat, S., Siriwan, W., Shanmugam, M.S., Viridis, S.G., 2024. Mapping and predicting cassava mosaic disease outbreaks using earth observation and meteorological data-driven approaches. *Remote Sensing Applications: Society and Environment* 35, 101231.
- Chikoti, P.C., Mulenga, R.M., Tembo, M., Sseruwagi, P., 2019. Cassava mosaic disease: a review of a threat to cassava production in zambia. *Journal of Plant Pathology* 101, 467–477.

- Effiong, B.U., 2021. Cassava Leaf Disease Classification with Deep Learning. Ph.D. thesis. AUST.
- Effiong, J.B., Enenyi, I.O., Etim, A.E., 2023. Perceived influence of agricultural extension on cassava production by the farmers in southern nigeria. *Agricultural Science Digest* 43, 883–888.
- Elfekih, S., Tay, W., Polaszek, A., Gordon, K., Kunz, D., Macfadyen, S., Walsh, T., Vyskočilová, S., Colvin, J., De Barro, P., 2021. On species delimitation, hybridization and population structure of cassava whitefly in africa. *Scientific Reports* 11, 7923.
- Eni, A.O., Efekemo, O.P., Onile-ere, O.A., Pita, J.S., 2020. South west and north central nigeria: Assessment of cassava mosaic disease and field status of african cassava mosaic virus .
- Eni, A.O., Efekemo, O.P., Onile-ere, O.A., Pita, J.S., 2021. South west and north central nigeria: Assessment of cassava mosaic disease and field status of african cassava mosaic virus and east african cassava mosaic virus. *Annals of Applied Biology* 178, 466–479.
- FAO, 2013. Food and agriculture organization of the united nations (fao). URL: <https://www.fao.org/home/en>, doi:10.15468/DL.RAYEKA.
- FAO, 2017. Food and agriculture organization of the united nations (fao). URL: <https://www.fao.org/home/en>, doi:10.15468/DL.RAYEKA.
- Farms, C., 2000. Disease control in cassava farms .
- Filani, M., Akintola, F., Ikporukpo, C., 1994. Ibadan region. R. Charles Publication.
- Herrera Campo, B.V., Hyman, G., Bellotti, A., 2011. Threats to cassava production: known and potential geographic distribution of four key biotic constraints. *Food Security* 3, 329–345.
- Kalischuk, M., Paret, M.L., Freeman, J.H., Raj, D., Da Silva, S., Eubanks, S., Wiggins, D., Lollar, M., Marois, J.J., Mellinger, H.C., et al., 2019. An improved crop scouting technique incorporating unmanned aerial vehicle–assisted multispectral crop imaging into conventional scouting practice for gummy stem blight in watermelon. *Plant disease* 103, 1642–1650.
- Kalyebi, A., Macfadyen, S., Hulthen, A., Ocitti, P., Jacomb, F., Tay, W.T., Colvin, J., De Barro, P., 2021. Within-season changes in land-use impact pest abundance in small-holder african cassava production systems. *Insects* 12, 269.
- Koima, I.N., 2018. Evaluating response to cassava viral disease infections among selected cassava genotypes under field and greenhouse assays in lower Eastern Kenya. Ph.D. thesis.

- Legg, J.P., Kumar, P.L., Makeshkumar, T., Tripathi, L., Ferguson, M., Kanju, E., Ntawurhunga, P., Cuellar, W., 2015. Cassava virus diseases: biology, epidemiology, and management, in: *Advances in virus research*. Elsevier. volume 91, pp. 85–142.
- Minato, N., Sok, S., Chen, S., Delaquis, E., Phirun, I., Le, V.X., Burra, D.D., Newby, J.C., Wyckhuys, K.A., de Haan, S., 2019. Surveillance for sri lankan cassava mosaic virus (slcmv) in cambodia and vietnam one year after its initial detection in a single plantation in 2015. *PLoS One* 14, e0212780.
- Modu, B., Polovina, N., Lan, Y., Konur, S., Asyhari, A.T., Peng, Y., 2017. Towards a predictive analytics-based intelligent malaria outbreak warning system. *Applied Sciences* 7, 836.
- Naeem, S., Ali, A., Anam, S., Ahmed, M.M., 2023. An unsupervised machine learning algorithms: Comprehensive review. *International Journal of Computing and Digital Systems* .
- Olasanmi, B., Kyallo, M., Yao, N., 2021. Marker-assisted selection complements phenotypic screening at seedling stage to identify cassava mosaic disease-resistant genotypes in african cassava populations. *Scientific Reports* 11, 2850.
- Olubanjo, O., 2016. Evaluation of quality of rainwater harvested from different roof catchments systems. *Journal of Sustainable Technology* 7.
- Regessa, D.T.B.M.D., 2017. Cassava integrated pest management: Review on cassava mosaic disease and mealybug. *Journal of Biology, Agriculture and Healthcare* 7, 1–7.
- Rey, C., Vanderschuren, H., 2017. Cassava mosaic and brown streak diseases: current perspectives and beyond. *Annual Review of Virology* 4, 429–452.
- Sambasivam, G., Opiyo, G.D., 2021. A predictive machine learning application in agriculture: Cassava disease detection and classification with imbalanced dataset using convolutional neural networks. *Egyptian informatics journal* 22, 27–34.
- Sastry, K.S., Sastry, K.S., 2013. Impact of virus and viroid diseases on crop yields. *Plant Virus and Viroid Diseases in the Tropics: Volume 1: Introduction of Plant Viruses and Sub-Viral Agents, Classification, Assessment of Loss, Transmission and Diagnosis* , 99–159.
- Soro, M., Tiendrébéogo, F., Pita, J.S., Traoré, E.T., Somé, K., Tibiri, E.B., Néya, J.B., Mutuku, J.M., Simporé, J., Koné, D., 2021. Epidemiological assessment of cassava mosaic disease in burkina faso. *Plant Pathology* 70, 2207–2216.
- Stephen, A., Akomolafe, P.O., Ogundoyin, K.I., 2021. A model for predicting malaria outbreak using machine learning technique. *Annals Computer Science Series* 19.

- Tembo, M., Mataa, M., Legg, J., Chikoti, P.C., Ntawuruhunga, P., 2017. Cassava mosaic disease: incidence and yield performance of cassava cultivars in zambia. *Journal of plant pathology* , 681–689.
- Ukoha, A.H., 2020. Assessment of the cost-benefits of climate change adaptation strategies of cassava-based farmers in southern nigeria. *International Journal of Environment and Climate Change* 10, 99–110.